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# Closing the Algebra Is Not Enough: Lie-Closure Regularization Makes Learned Transformation Structure Comparable, Not Identifiable

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## Abstract

When do independently trained models learn the same *transformation structure*, and when does agreement between them reflect the truth? We study this for transition-based representation learners that infer latent actions from pairs of observations and represent them with learned skew-symmetric generators. We add a Lie-closure penalty (BracketNet) and evaluate on  $\text{SO}(2)$ ,  $T^2$  and  $\text{SO}(3)$  actions observed through nonlinear maps and through rendered images, with every choice fixed on development seeds and fresh seeds evaluated once.

(1) Transition fitting alone recovers the abelian  $T^2$  algebra in most runs, but leaves the  $\text{SO}(3)$  generator span non-closed in 15/20 runs. Closure regularization closes it, roughly halves the distance to the true algebra, and makes the algebras of independently trained models agree more (Holm-corrected  $p \leq 8.8 \times 10^{-3}$ ), with no detectable change in transport error. (2) Closed is not correct. 8/20  $\text{SO}(3)$  runs close onto a *chiral*  $\mathfrak{su}(2)$ , and on rendered images every closed run (7/20) is chiral. We prove a partial explanation: a rotation about the axis through a point leaves that point fixed, so the true action cannot be inferred from endpoints and no continuous endpoint rule makes the true algebra exact, whereas a chiral algebra acts simply transitively and admits an exact rule. Composition errors, a pre-registered intervention with true action coefficients, and a generator-span trace agree with this account. (3) Transfer of transformations between models tracks whether they learned the *same* algebra, even a wrong one. Linear CKA catches encoder failures but not these class mismatches.

Algebraic closure turns continuous disagreement between models into disagreement among a few closed subalgebras: it makes learned transformation structure comparable, but not identifiable.

## 1 Introduction

Representation-similarity methods ask whether two networks embed data in the same way, up to a class of transformations [Kornblith et al., 2019, Williams et al., 2021, Moschella et al., 2023]. For models that learn how data *transform*, there is a second question: do they agree on the transformations themselves? This matters when actions inferred by one model are to be used in another, and when a learned symmetry is read as a scientific statement. Embedding agreement does not settle it. Two

31 models can embed the same data almost identically and still represent its transformations by different  
32 Lie algebras, and two models can agree perfectly with each other while both being wrong (Cor. 1).

33 We study this in transition-based symmetry learning [Park et al., 2022, Keurti et al., 2023, Yang et al.,  
34 2024]. A model infers a latent action from a pair of observations and represents it through learned  
35 generators. Nothing in a transition loss requires those generators to span a Lie algebra, so a natural  
36 structural bias is to penalize commutators that leave their span. We ask: *when do independently*  
37 *learned transformation representations become structurally comparable, and why does algebraic*  
38 *consistency fail to make them identifiable?*

## 39 Contributions.

- 40 • **An evaluation that separates five levels of agreement:** embedding, algebra, representation  
41 class, inferred action and ground truth. It includes cross-model transformation transfer, uses  
42 disjoint seed pairs as independent units, and pre-registers its primary tests.
- 43 • **Evidence that closure changes the *kind* of disagreement.** For  $SO(3)$ , transition fitting  
44 leaves generator spans non-closed and scattered. A closure penalty concentrates them  
45 on the three closed classes allowed in  $\mathfrak{so}(4)$ , one correct and two chiral, which improves  
46 cross-model agreement without guaranteeing correctness.
- 47 • **A mechanism with a proof (Prop. 3).** Endpoint-based action inference cannot capture  
48 rotations about the axis through a point, and no continuous endpoint rule makes the true  
49 algebra exact, whereas a chiral algebra admits an exact rule. The result is qualitative (no  
50 quantitative gap). Three tests, two of them pre-registered, support this account.
- 51 • **A transfer analysis:** actions transfer between models that learned the same algebra, right or  
52 wrong. CKA catches representation failures but not algebra-class mismatches.

53 We also give a corrected invariance analysis of the closure loss and a local bound on how far products  
54 of group elements leave the learned span. The underlying Lie theory is standard; we cite it as such.

## 55 2 Related work

56 **Learning transformation groups.** Lie-group transformation models have a long history [Rao  
57 and Ruderman, 1999, Sohl-Dickstein et al., 2010, Cohen and Welling, 2014], including transport  
58 operators in autoencoder latents [Connor and Rozell, 2020]. Recent methods discover symmetries  
59 from predictors [Dehmamy et al., 2021], adversarially from distributional invariance [Yang et al.,  
60 2023], in learned latent spaces [Yang et al., 2024], beyond affine families [Shaw et al., 2024], or as  
61 one-parameter families [Gabel et al., 2023]. Closure losses that force learned generators to span a  
62 (sub)algebra already exist [Forestano et al., 2023], including studies of  $\mathfrak{so}(4)$  subalgebras; our closure  
63 term is an instance of this idea. Equivariant networks assume a known group [Cohen and Welling,  
64 2016, Bronstein et al., 2021].

65 **Group-structured representations from transitions.** Symmetric embedding networks [Park et al.,  
66 2022] use known actions. Homomorphism autoencoders [Keurti et al., 2023] learn a homomorphism  
67 from action labels into a matrix group without knowing the group. Symmetry-based disentanglement  
68 [Higgins et al., 2018] defines representations through group actions and needs interaction [Caselles-  
69 Dupré et al., 2019]. Rotation representations of an agent’s actions have been learned from interaction  
70 sequences [Quessard et al., 2020], and actions have been estimated without labels [Painter et al.,  
71 2020], and group decompositions have been shown identifiable under assumptions [Dang-Nhu et al.,  
72 2026]. We share the transition setting and, like Painter et al. [2020], use *no* action labels. Our  
73 oracle-coefficient intervention (Sec. 6.2) is the action-supervised counterpart and shows what the  
74 missing labels cost. LieGAN and LaLiGAN learn from distributional signals rather than transitions,  
75 so we do not compare against them directly.

76 **Comparing representations.** CKA [Kornblith et al., 2019], shape metrics [Williams et al., 2021]  
77 and relative representations [Moschella et al., 2023] compare embeddings up to a transformation  
78 class, and convergence of representations across models is an active hypothesis [Huh et al., 2024].  
79 These tools compare *point clouds*. We compare the transformation algebras acting on them, and the  
80 transfer of actions between models.

81 **What is not new.** Closure penalties, the classification of subalgebras of  $\mathfrak{so}(4)$  and  $\mathfrak{so}(5)$ , Dynkin’s  
82 formula, the hairy-ball theorem and quaternion transitivity are standard. Topological obstructions to  
83 continuous encoders and rotation representations are known [de Haan and Falorsi, 2018, Zhou et al.,  
84 2019, Bouchacourt et al., 2021]; Prop. 3b is an obstruction of the same kind, applied to endpoint  
85 action inference. Our contribution is their consequences for this learning problem and the empirical  
86 analysis. Appendix D compares settings in detail.

### 87 3 Setting and method

88 Observations  $x = h(u)$  come from states  $u$  through an unknown injective map. Training data are  
89 unlabeled triples  $(x_0, x_1, x_2) = (h(u), h(g_1 u), h(g_2 g_1 u))$  with near-identity  $g_i$  from an unknown  
90 compact group, and  $K$  is given.

91 The model has an encoder  $E$ , a decoder  $D$ , an action-inference network  $a = q(z, z')$  (bounded  
92 by  $0.65 \tanh$ ) and generators  $A_k \in \mathfrak{so}(d)$ , each rescaled to  $\|A_k\|_F = \sqrt{2}$ . An action is  $\rho(a) =$   
93  $\exp(\sum_k a_k A_k)$ .

94 With  $z_t = E(x_t)$  and  $a_{st} = q(z_s, z_t)$ , the losses are:

- 95 • reconstruction  $\sum_t \|D(z_t) - x_t\|^2$ ;
- 96 • transport in latent and observation space,  $\sum_{t < 2} \|\rho(a_{t,t+1})z_t - z_{t+1}\|^2$  and  
97  $\sum_{t < 2} \|D(\rho(a_{t,t+1})z_t) - x_{t+1}\|^2$ ;
- 98 • composition  $\|\rho(a_{02}) - \rho(a_{12})\rho(a_{01})\|_F^2$ ;
- 99 • a Gram penalty  $\|B^\top B - 2I\|_F^2$  and an anti-collapse term  $\|\text{Cov}(z) - I\|_F^2$ ;
- 100 • the closure residual

$$\mathcal{L}_{\text{close}} = \frac{1}{K^2} \sum_{i,j} \|(I - P_\varepsilon) \text{vec}[A_i, A_j]\|_2^2, \quad P_\varepsilon = B(B^\top B + \varepsilon I)^{-1} B^\top, \quad (1)$$

101 where  $B$  stacks the vectorized generators.

102 LOCAL uses reconstruction, transport and the regularizers. +COMP. adds composition (weight 1.2).  
103 BRACKETNET adds  $\lambda(t)\mathcal{L}_{\text{close}}$ :  $\lambda(t) = 0$  for 35% of updates, a linear ramp to  $\lambda$  by 70%, and frozen  
104 generators afterwards. All three share networks, data and optimizer (weights in Appendix B).

### 105 4 Theory: what closure guarantees, and why it can prefer the wrong algebra

106 Notation:  $\mathfrak{a} = \text{span}(A)$ ,  $G = B^\top B$ ,  $P_0$  the orthogonal projector onto  $\mathfrak{a}$ ,  $N = I - P_0$ . Proofs are in  
107 Appendix A. Standard Lie-theoretic facts are cited as such; our contributions are their consequences  
108 for this learning objective.

109 **Invariances.** With  $\varepsilon=0$  and  $M \in GL(K)$ ,  $\mathcal{L}_{\text{close}}(BM) = K^{-2} \|R(M \otimes M)\|_F^2$ , where  $R$  is the  
110 residual matrix, so  $\sigma_{\min}(M)^4 \mathcal{L}_{\text{close}} \leq \mathcal{L}_{\text{close}}(BM) \leq \sigma_{\max}(M)^4 \mathcal{L}_{\text{close}}$ . The loss is invariant only  
111 under orthogonal reparameterizations; its zero set (“ $\mathfrak{a}$  is a subalgebra”) is  $GL(K)$ -invariant. For  
112  $\varepsilon > 0$ ,  $P_\varepsilon(BM) = B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$ . Because  $\text{diag } G = 2$ , the Gram target affects no  
113 gradient.

114 **Proposition 1** (Closure, escape and integration). (a) If  $A_1, \dots, A_K$  are independent and  $[A_i, A_j] \in \mathfrak{a}$ ,  
115 then  $\mathfrak{a}$  is a Lie subalgebra and integrates to a connected immersed subgroup of  $SO(d)$  (stan-  
116 dard). (b) The uniform escape constant  $\delta = \sup_{X,Y \in \mathfrak{a}} \|N[X, Y]\|_F / (\|X\|_F \|Y\|_F)$  satisfies  $\delta \leq$

117  $K\sqrt{\mathcal{L}_{\text{close}}(A; 0)/\lambda_{\min}(G)}$ . (c) For  $X, Y \in \mathfrak{a}$  with  $\|X\|_F, \|Y\|_F \leq r < \frac{\ln 2}{4}$ ,  $\text{dist}(\log(e^X e^Y), \mathfrak{a}) \leq$   
118  $\frac{\delta}{4}[-\ln(2 - e^{4r}) - 4r] = \frac{1}{2}\delta r^2 + O(\delta r^3)$ . (d) Chained orthogonal transport accumulates per-step  
119 errors at most additively.

120 Part (c) is local: it covers rotations of up to  $\approx 0.12$  rad, smaller than typical inferred actions.

121 **Closed subalgebras in the benchmarks.** Write  $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$  (two commuting ideals).

122 **Proposition 2** (Classification; standard facts applied to our benchmarks). (a) Every 2-dimensional  
123 subalgebra of  $\mathfrak{so}(5)$  is abelian and  $SO(5)$ -conjugate to the true  $\mathfrak{t}^2$ . (b) Every 3-dimensional subalgebra  
124 of  $\mathfrak{so}(4)$  is isomorphic to  $\mathfrak{su}(2)$ . It is  $SO(4)$ -conjugate to exactly one of three: the diagonal  
125  $\mathfrak{so}(3)$  acting on  $\mathbb{R}^3 \oplus \mathbb{R}$  (the truth),  $\mathfrak{su}(2)_L$  or  $\mathfrak{su}(2)_R$ . Under  $O(4)$  the two chiral factors merge into  
126 one class. (c) No closed 4-dimensional span in  $\mathfrak{so}(4)$  contains the diagonal  $\mathfrak{so}(3)$ .

127 The three  $SO(3)$  classes are separated by the Pfaffian of unit elements  $(0, +\frac{1}{2}, -\frac{1}{2})$  and by the  
128 participation ratio (2 vs. 4). Diagonal and chiral spans are at subspace distance exactly  $\frac{1}{2}$  (principal  
129 angles of  $45^\circ$ ). An optimization that does not use the classification confirms it: minimizing a span-  
130 only closure residual from 200 random 3-dimensional subspaces of  $\mathfrak{so}(4)$  converged to 192 diagonal  
131 and 8 chiral spans, and to nothing else.

132 **Endpoint ambiguity: why the objective can favour a wrong closed algebra.** BracketNet infers  
133 each action from the two endpoint codes,  $a = q(z, z')$ . For the true  $SO(3)$  action this is ill-posed.  
134 A rotation about the axis through  $z$  fixes  $z$ , so  $g$  and  $g \exp(t\hat{z})$  have the same endpoints. A chiral  
135  $\mathfrak{su}(2)$  has no such ambiguity: left multiplication by unit quaternions acts simply transitively on every  
136 sphere of  $\mathbb{R}^4 \cong \mathbb{H}$ . Call an endpoint rule  $q$  exact on a triple  $(z_0, z_1, z_2)$  if  $\rho(q(z_i, z_j))z_i = z_j$  for  
137  $ij \in \{01, 12, 02\}$  and  $\rho(q(z_0, z_2)) = \rho(q(z_1, z_2))\rho(q(z_0, z_1))$ .

138 **Proposition 3** (Endpoint ambiguity). On the  $SO(3)$  benchmark, use the ground-truth state as the  
139 latent code, with an exact decoder. (a) With chiral generators,  $q(z, z') = \log(z' z^{-1})$  is exact on  
140 every triple on a sphere  $\|z\| = \text{const} > 0$  (all data triples); it is smooth wherever  $z' \neq -z$ . The  
141 closure, Gram and covariance terms vanish too. Exactness fails only where a coefficient exceeds  
142 the bound. (b) With the true generators, for every  $\epsilon > 0$  and every orbit sphere  $\{(v, z_4) : \|v\| = c\}$ ,  
143  $c > 0$ , no continuous rule is exact on all triples in that sphere with pairwise angles  $< \epsilon$ . (c) For  
144 any action-inference rule based on endpoints,  $\mathbb{E}\|\rho(q(z, gz)) - g\|^2 > 0$  under the true action: the  
145 stabilizer component of  $g$  is not identifiable from  $(z, gz)$ .

146 Part (b) follows from the hairy-ball theorem: exactness on small triangles would define a continuous  
147 section of  $SO(3) \rightarrow S^2$ . It concerns exact zero for continuous rules only. A discontinuous rule can  
148 be exact off a null set, and we do not bound how close continuous rules can come (App. A, Remark).  
149 On the data, ignoring the coefficient bound, the minimal-rotation rule for the true algebra transports  
150 exactly but leaves a composition residual of  $3.2 \times 10^{-3}$ ; the chiral rule leaves  $\sim 10^{-31}$ .

151 **Corollary 1** (Agreement without correctness). Two models with identical ground-truth embeddings  
152 and  $\mathfrak{su}(2)_L$  generators have linear CKA 1 with each other and with the true state, zero closure,  
153 transport and composition loss (up to coefficient clipping), and cross-model algebra distance 0, but  
154 distance  $\frac{1}{2}$  to the true algebra and inferred actions that are not the true rotations.

155 **The implication chain, made precise.** Write (i) closure, (ii) correct conjugacy class, (iii) recovery  
156 after latent alignment, (iv) correct action inference. (i) $\Rightarrow$ (ii) holds for  $T^2$  (Prop. 2a) and fails for  
157  $SO(3)$  (Prop. 2b). (ii) $\Rightarrow$ (iii) also needs an encoder close to an isometry. (iv) is unattainable for  
158  $SO(3)$  by every endpoint method, even with the true algebra (Prop. 3c). Agreement between models  
159 implies none of (ii)–(iv) (Cor. 1). Proposition 3 describes the loss landscape at one embedding, and  
160 qualitatively. It does not show that the minimum over trainable rules is lower for chiral solutions, nor  
161 which solution gradient descent reaches; Sec. 6.2 tests the latter empirically.

## 5 Experimental protocol

**Benchmarks.**  $\text{SO}(2)$ ,  $T^2$  and  $\text{SO}(3)$  act linearly on a latent state in  $\mathbb{R}^d$  that includes an invariant coordinate ( $d = 3, 5, 4$ ). The true algebra is therefore a proper subspace of  $\mathfrak{so}(d)$ , and random spans do not close: the mean closure of random normalized spans is 0.60 ( $K=2$  in  $\mathfrak{so}(5)$ ) and 0.66 ( $K=3$  in  $\mathfrak{so}(4)$ ).

States are drawn as  $z_0 \sim \mathcal{N}(0, I)$ , and transitions apply  $\exp(\sum_k a_k T_k)$  with  $a_k \sim \mathcal{N}(0, 0.24^2)$ . Observations are  $x = Qz + 0.18 \tanh(Bz)$ , one fixed injective map per group. The rendered benchmark is described in Sec. 6.3. Each run trains on 1,400 triples and is evaluated on 500 ten-edge sequences.

**Selection on development seeds only.** We fixed the training budget, closure weight and closure objective on development seeds 5–9, with rules committed before those runs:

- *Budget*: the smallest budget at which +COMP. transport is within 10% of its value at 4,000 updates. This rule is method-agnostic.
- *Closure weight*: the  $\lambda$  with the lowest closure among those whose transport is not worse than +COMP.'s. This rule needed an amendment, committed before any fresh run: no  $\lambda$  met the constraint, so the tolerance was relaxed in steps of 10%, and  $\lambda = 3$  qualified at 1.20.

The result is 4000 updates,  $\lambda = 3$  and Eq. (1). All runs evaluate the final iterate.  $\lambda = 0.1$ , the originally planned weight, is evaluated at the same budget as a reference.

**Seeds and pre-registration.** Main evaluation: 20 fresh seeds (100–119). Ablations: 200–209. Rendered benchmark: 300–319. Oracle intervention: 400–419. Each set was used once. Structural thresholds, primary hypotheses and the predictions for the mechanism and transfer experiments were committed before the corresponding runs.

**Structural categories (fixed in advance).** A run is *collapsed* if  $\sigma_{\min}/\sigma_{\max}(B) < 0.1$ , and *closed* if its closure is at most 1% of the random-span level. It is *correct-closed* if it is closed and its ground-truth algebra distance (mean squared sine of the principal angles between  $\text{span}(RAR^T)$  and the true span, with  $R$  the Procrustes map from codes to true states) is at most 10% of chance. A closed  $\text{SO}(3)$  span with participation ratio  $\geq 3$  is *chiral*.

**Statistics.** Seeds are the independent units; cross-seed quantities use the 10 *disjoint* seed pairs. The primary family compares BracketNet with +COMP., one-sided, with exact paired sign-flip tests and Holm correction: closure (H1), ground-truth algebra distance (H2) and cross-seed algebra distance (H3), each for  $T^2$  and  $\text{SO}(3)$ . Everything else is secondary, and we call a difference detectable only if its 95% interval excludes zero.

## 6 Results

### 6.1 Closure makes $\text{SO}(3)$ algebras closed, but often wrong

**Transition fitting closes  $T^2$  but not  $\text{SO}(3)$ .** Without closure regularization, the true  $T^2$  algebra is recovered in 14/20 (+Comp.) and 14/20 (Local) runs, while the  $\text{SO}(3)$  span stays *non-closed* in 15/20 and 14/20 runs. One- and ten-step transport errors are low in both cases (Table 1). BracketNet at the originally planned  $\lambda=0.1$  behaves like the baselines.

**Closure regularization: large effects for  $\text{SO}(3)$ , concentrated effects for  $T^2$ .** For  $\text{SO}(3)$ , closure falls from 0.6163 to 0.0250 (lower on 20/20 seeds), and the ground-truth distance falls from 0.4754 to 0.2429 (19/20). Both are significant after Holm correction (Table 2).

For  $T^2$  the closure test is also significant, but the effect is concentrated: the median paired difference is  $-8.9 \times 10^{-5}$ , and the differences are heavily skewed (skewness -1.94). The  $t$ -interval includes

Table 1: Fresh-seed results (seeds 100–119, mean $\pm$ s.d.; frozen protocol). GT dist.: ground-truth algebra distance after Procrustes alignment (chance 0.80 for  $T^2$ , 0.50 for SO(3)). CC: correct-closed runs. SO(2), where all methods coincide, is in Appendix Table 4.

| Group | Method                       | 1-step MSE          | 10-step MSE         | Closure residual                            | GT dist.          | CC    |
|-------|------------------------------|---------------------|---------------------|---------------------------------------------|-------------------|-------|
| $T^2$ | Local                        | 0.0080 $\pm$ 0.0076 | 0.0645 $\pm$ 0.0841 | $3.4 \times 10^{-1} \pm 7.2 \times 10^{-1}$ | 0.161 $\pm$ 0.282 | 14/20 |
| $T^2$ | +Comp.                       | 0.0081 $\pm$ 0.0076 | 0.0605 $\pm$ 0.0757 | $2.9 \times 10^{-1} \pm 6.3 \times 10^{-1}$ | 0.151 $\pm$ 0.274 | 14/20 |
| $T^2$ | BracketNet                   | 0.0072 $\pm$ 0.0079 | 0.0539 $\pm$ 0.0792 | $2.2 \times 10^{-2} \pm 5.6 \times 10^{-2}$ | 0.129 $\pm$ 0.282 | 16/20 |
| $T^2$ | BracketNet ( $\lambda=0.1$ ) | 0.0082 $\pm$ 0.0078 | 0.0622 $\pm$ 0.0785 | $3.1 \times 10^{-1} \pm 6.1 \times 10^{-1}$ | 0.157 $\pm$ 0.277 | 14/20 |
| SO(3) | Local                        | 0.0048 $\pm$ 0.0027 | 0.0325 $\pm$ 0.0312 | $6.3 \times 10^{-1} \pm 5.7 \times 10^{-1}$ | 0.479 $\pm$ 0.435 | 6/20  |
| SO(3) | +Comp.                       | 0.0048 $\pm$ 0.0031 | 0.0328 $\pm$ 0.0299 | $6.2 \times 10^{-1} \pm 5.7 \times 10^{-1}$ | 0.475 $\pm$ 0.433 | 5/20  |
| SO(3) | BracketNet                   | 0.0049 $\pm$ 0.0034 | 0.0319 $\pm$ 0.0341 | $2.5 \times 10^{-2} \pm 7.1 \times 10^{-2}$ | 0.243 $\pm$ 0.254 | 8/20  |
| SO(3) | BracketNet ( $\lambda=0.1$ ) | 0.0050 $\pm$ 0.0031 | 0.0484 $\pm$ 0.0484 | $5.4 \times 10^{-1} \pm 5.4 \times 10^{-1}$ | 0.453 $\pm$ 0.413 | 5/20  |

Table 2: Pre-registered primary hypotheses, BracketNet vs. +COMP. (one-sided). Units: seeds ( $n=20$ ) for H1–H2, disjoint seed pairs ( $n=10$ ) for H3. Reported: two-sided 95%  $t$ -interval of the difference, paired effect size  $d_z$ , and the number of units where BracketNet is lower.  $p$ : exact sign-flip; Holm over six tests.

| Hypothesis               | Group | $n$ | +Comp. | BracketNet | 95% CI ( $t$ )     | $d_z$ | lower | $p$     | Holm $p$ |
|--------------------------|-------|-----|--------|------------|--------------------|-------|-------|---------|----------|
| H1 closure               | $T^2$ | 20  | 0.2906 | 0.0216     | [-0.5427, +0.0049] | -0.46 | 16/20 | 4.3e-05 | 2.1e-04  |
| H2 ground-truth distance | $T^2$ | 20  | 0.1506 | 0.1289     | [-0.0674, +0.0240] | -0.22 | 16/20 | 1.6e-01 | 3.1e-01  |
| H3 cross-seed distance   | $T^2$ | 10  | 0.2138 | 0.1981     | [-0.1421, +0.1106] | -0.09 | 7/10  | 3.8e-01 | 3.8e-01  |
| H1 closure               | SO(3) | 20  | 0.6163 | 0.0250     | [-0.8550, -0.3278] | -1.05 | 20/20 | 9.5e-07 | 5.7e-06  |
| H2 ground-truth distance | SO(3) | 20  | 0.4754 | 0.2429     | [-0.3348, -0.1301] | -1.06 | 19/20 | 1.2e-04 | 4.9e-04  |
| H3 cross-seed distance   | SO(3) | 10  | 0.6845 | 0.3922     | [-0.4424, -0.1423] | -1.39 | 9/10  | 2.9e-03 | 8.8e-03  |

zero, while the interval obtained by inverting the primary sign-flip test,  $[-0.5331, -0.0067]$ , does not. The difference comes from the few seeds in which +COMP. fails to close. The  $T^2$  ground-truth distance does not change significantly.

Ten-step transport shows no detectable difference in either group; the 95% intervals are  $[-0.0199, 0.0066]$  and  $[-0.0080, 0.0061]$ .

**Three SO(3) outcomes.** Every run falls into one of the classes of Prop. 2 or is non-closed (Fig. 1a).

- BracketNet: 8 correct (diagonal), 8 chiral, 4 non-closed.
- +COMP.: 5 correct, 0 chiral, 15 non-closed.

Closure converts non-closed runs into closed ones, and about half of them land in the wrong class. The rise in correct-closed runs is not significant (exact McNemar  $p = 0.25$ ).

## 6.2 Mechanism: the endpoint objective favours chiral solutions

We tested Prop. 3 with predictions registered before any of the following runs.

**Composition error by class (existing runs).** As predicted, chiral runs compose far better than correct ones: median composition error  $3.3 \times 10^{-5}$  (chiral) vs.  $7.9 \times 10^{-4}$  (correct). Their one-step errors are the same (0.0035 vs. 0.0034). Their action disagreement with the truth is 1.01 vs. 0.38. The correct runs' value is the floor imposed by Prop. 3(c); chiral runs are no better than the identity.

**Closure geometry alone favours the correct class.** Pure closure flow from random subspaces reaches the diagonal class in 192/200 starts (Sec. 4), so the high chiral rate in training is not a property of the closure penalty alone.

**Trace of the generator span.** We re-ran the 20 BracketNet SO(3) runs with generator snapshots; 20/20 reproduced the stored runs bit-for-bit. We measured each span's chirality  $\chi = e_L - e_R$ , the share of its energy in  $\mathfrak{su}(2)_L$  minus that in  $\mathfrak{su}(2)_R$ .

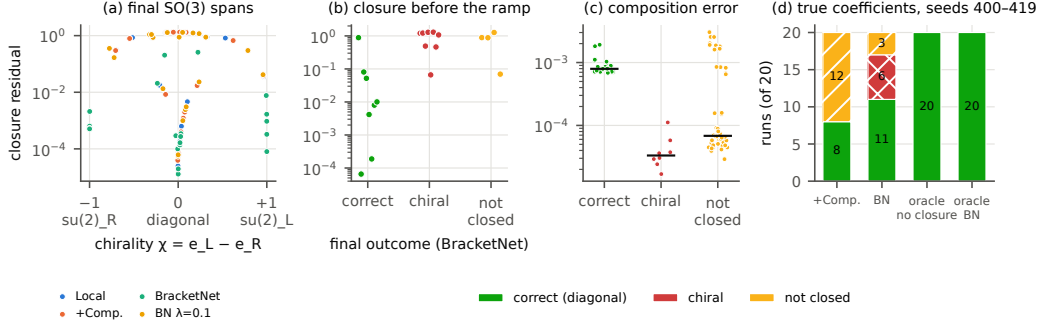


Figure 1: (a) Final SO(3) span chirality  $\chi$  vs. closure residual for every fresh-seed run: closed runs sit at  $\chi \in \{-1, 0, 1\}$ , the classes of Prop. 2. (b) Closure of the BracketNet span when the closure penalty switches on, by final outcome (post hoc). (c) Composition error by outcome class, all SO(3) runs. (d) Outcomes with endpoint-inferred vs. true (oracle) action coefficients, seeds 400–419.

- *Pre-registered.* At the start of the closure ramp,  $|\chi|$  predicts a chiral outcome among the 16 closed runs with AUC 0.75 (at initialization: 0.61). Medians are 0.22 (ending chiral) vs. 0.13 (ending correct).
- *Post hoc.* The closure residual at the ramp start separates outcomes more sharply (AUC 0.94). 7 of the 8 runs that end correct had already nearly closed onto the diagonal during the transition-fitting phase. 7 of the 8 chiral runs were still far from closed when the penalty switched on (Fig. 1b).

Together with the closure flow, this suggests the following picture. Transition fitting sometimes finds the diagonal algebra by itself, and closure then keeps it. When the span is still far from closed, closing it under the transport and composition losses, which admit exact chiral solutions at the ground-truth embedding but no exact continuous true-algebra solution (Prop. 3), usually yields a chiral algebra.

**Intervention: supplying the true action coefficients (pre-registered).** Prop. 3 predicts that chiral solutions lose their advantage if actions need not be inferred from endpoints. On 20 new seeds (400–419) we trained BracketNet with the true coefficients in the transport terms. This is the action-supervised setting of homomorphism autoencoders [Keurti et al., 2023];  $q$  is fit only for evaluation. Results, relative to standard BracketNet on the same seeds:

- chiral outcomes fell from 6/20 to 0/20 (exact McNemar  $p = 0.03$ );
- correct-closed runs rose from 11/20 to 20/20 ( $p = 3.9 \times 10^{-3}$ );
- the ground-truth distance changed by  $-0.215$  ( $[-0.332, -0.098]$ );
- without any closure term, the oracle model also reaches 20/20 correct-closed runs.

Once actions are given, closure regularization is unnecessary: the difficulty lies in inferring actions from endpoints. The oracle models’ own  $q$ , which must infer actions from endpoints at test time, still has action disagreement 0.41, the floor of Prop. 3(c). The intervention adds information as well as removing the ambiguity, so it supports the mechanism but does not isolate it.

### 6.3 A rendered-image benchmark

To test a harder observation process, we rendered the SO(3) state as  $20 \times 20$  images: the 3-D position of a Gaussian blob sets its image position and size, and the invariant coordinate sets its brightness ( $p = 400$ ). The frozen protocol was applied unchanged on 20 new seeds (300–319), with predictions registered in advance.

- *No baseline run closes* (Local 20/20, +COMP. 20/20 non-closed). BracketNet lowers closure on 20/20 seeds (0.062 vs. 1.227, one-sided  $p = 9.5 \times 10^{-7}$ ).

- *Every closed BracketNet run is chiral* (7/20; none diagonal). This is read off the chirality invariant, which needs no latent alignment.
- *Ground-truth recovery cannot be assessed by our alignment metric here.* The learned latents are far from an isometry of the true state (CKA with it 0.623, Procrustes residual 0.611).
- *There is a small transport cost:* 10-step error rises by  $4.5 \times 10^{-4}$  ( $[1.4 \times 10^{-4}, 7.7 \times 10^{-4}]$ ).

Under rendering, closure regularization therefore produced closed algebras only of the wrong class. The benchmark is still synthetic and does not test natural images.

## 7 Representation alignment: agreement in embeddings vs. agreement in transformations

Models trained from different seeds can be compared at five levels: embedding geometry, transformation algebra, group representation, inferred action, and ground truth. We compare each pair of models on a shared probe set:

- *alignment*: the orthogonal Procrustes map  $R$  from  $E_a(x)$  to  $E_b(x)$ , fitted on probe set A;
- *embedding agreement*: linear CKA;
- *algebra agreement*: the cross-seed algebra distance between  $\text{span}(RA^{(a)}R^\top)$  and  $\text{span}(A^{(b)})$ ;
- *transfer*: on a held-out probe set B, apply the action inferred by  $a$  in  $b$ 's space,  $\hat{x}_{t+1} = D_b(R\rho_a R^\top E_b(x_t))$ . The *transfer gap* is 0 when this is as good as  $b$ 's own action and 1 when it is no better than the identity.

Independent units are 10 disjoint seed pairs, the same for every method (Appendix Table 5).

**Algebras agree more, through a few discrete classes.** For  $\text{SO}(3)$ , the cross-seed algebra distance falls from 0.6845 to 0.3922 (lower in 9/10 pairs, Holm  $p = 8.8 \times 10^{-3}$ ). For  $T^2$  it does not change significantly.

The  $\text{SO}(3)$  pairs separate by class:

- both models correct: distance 0 (1 pair);
- both in the same *wrong* chiral class: distance 0 (1 pair);
- correct vs. chiral: distance  $\approx \frac{1}{2}$ , as fixed by Prop. 2 (4 pairs);
- at least one model non-closed: 4 pairs.

All 10 +COMP. pairs involve a non-closed model and spread over  $[0, 1]$ . Closure thus replaces continuous disagreement with disagreement between a few closed classes, so cross-model agreement does not certify correctness (Cor. 1).

**Transfer tracks structural agreement, not correctness.** Over all 190 pairs of each method (pooled, descriptive):

- $\text{SO}(3)$  pairs of two correct models transfer almost perfectly: median gap 0.013, 0% failures (gap  $> 0.5$ ), over 63 pairs.
- Pairs of two models in the same wrong class also transfer: median gap 0.177, 0% failures. Aligning them through the ground truth instead of to each other breaks transfer (median gap 4.151).
- Correct-vs-chiral pairs transfer worse: median gap 0.365, 19% failures.

What makes transformations reusable across models is that both models learned the *same* algebra, whether or not it is correct.



**CKA vs. structural distance as predictors of transfer failure (pre-registered; prediction not supported).** We pre-registered that the label-free algebra distance would predict transfer failure better than CKA. It did not.

- AUC for  $T^2$ : CKA 1.00 vs. algebra distance 0.98.
- AUC for SO(3): CKA 0.75 vs. algebra distance 0.62; difference interval  $[-0.40, 0.30]$  (seed-level cluster bootstrap).

Most transfer failures come from seed-level encoder failures, which CKA detects. CKA is, however, blind to *class* mismatches: SO(3) correct-vs-chiral pairs have median CKA 0.989, the same as correct pairs (0.991), yet a far larger transfer gap. Embedding similarity and algebra agreement capture different failure modes, and a structural comparison needs both.

## 8 Ablations

Ablations use seeds 200–209 with same-seed references (Appendix Table 7).

- *Replication.* +COMP. again has higher SO(3) closure and ground-truth distance than Bracket-Net (+0.2176,  $[0.0786, 0.3567]$ ).
- *Anti-collapse term.* Essential: at weight 1 the  $T^2$  latent distorts (CKA with the truth 0.73) and correct-closed runs fall to 0/10.
- *No detectable effect:* the fit–close–refit schedule (constant penalty or no freeze), the Gram penalty, and a span-only closure objective. We do not claim the schedule is necessary.
- *Misspecified  $K$ .* With  $K+1$  no run is correct-closed (0/10 for SO(3)), as Prop. 2(c) requires.

## 9 Limitations

- **Synthetic scope.** Both benchmarks, including the rendered one, are synthetic, low-dimensional and compact, with known  $K$  and orthogonal actions. We make no claim about natural images, physical systems or large models.
- **No identifiability.** Wrong closed algebras are frequent for SO(3), and correct algebras still have irreducible action ambiguity (Prop. 3c). Our ground-truth comparisons use orthogonal Procrustes, so they assume a near-isometric encoder; the  $T^2$  failures are seed-level fitting failures shared by all methods.
- **Limited baselines.** We compare controlled objective variants and an action-supervised oracle; we do not reimplement published methods with different supervision.
- **Statistics and protocol.** We use 20 seeds; effects outside SO(3) are small or concentrated in few seeds. No transport improvement is established; on rendered images transport is slightly worse. The closure-weight rule needed an amendment, committed before any fresh run. One pre-registered prediction (structure beats CKA at predicting transfer failure) was not supported.
- **Theory is local, qualitative and self-checked.** The escape bound covers only small actions; the classification is specific to  $\mathfrak{so}(4)$  and  $\mathfrak{so}(5)$ ; Prop. 3 concerns exact zeros at the ground-truth embedding, gives no quantitative gap and says nothing about the optimization path. The proofs have not been checked independently of the AI system that drafted them.

## 10 Conclusion

Closure turns continuous disagreement between transformation models into a choice among a few closed subalgebras. For SO(3) the choice is often wrong, because endpoints cannot reveal rotations about the axis through a point. Agreement, of embeddings or of transformations, is not evidence of correctness.

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## 413 A Proofs

414 **Invariance law.** Under  $B \mapsto BM$ ,  $[A'_i, A'_j] = \sum_{k,l} M_{ki} M_{lj} [A_k, A_l]$ , so the commutator matrix  
 415 becomes  $C(M \otimes M)$ , while  $P_0(BM) = P_0(B)$ . Hence  $R \mapsto R(M \otimes M)$ , and the bounds  
 416 follow from  $\sigma(M \otimes M) = \{\sigma_i \sigma_j\}$ . For  $\varepsilon > 0$ ,  $P_\varepsilon(BM) = BM(M^\top GM + \varepsilon I)^{-1} M^\top B^\top =$   
 417  $B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$ .

418 **Proposition 1.** (a) The commutator is bilinear and antisymmetric and satisfies Jacobi, so closure  
 419 makes  $\mathfrak{a}$  a Lie subalgebra; the Lie subgroup theorem [Hall, 2015, Ch. 5] gives the integrating subgroup.  
 420 It need not be closed: an irrational line in a torus is an example.

421 (b) Write  $X = \sum x_i A_i$ ,  $Y = \sum y_j A_j$  and  $r_{ij} = N[A_i, A_j]$ . Then  $\|N[X, Y]\| \leq$   
 422  $\|x\| \|y\| (\sum \|r_{ij}\|^2)^{1/2} = \|x\| \|y\| K \sqrt{\mathcal{L}_{\text{close}}}$  by Cauchy–Schwarz, and  $\|X\|^2 = x^\top G x \geq \lambda_{\min} \|x\|^2$ .

423 (c) Dynkin’s form [Bonfiglioli and Fulci, 2012] is  $\log(e^X e^Y) = \sum_{m \geq 1} \frac{(-1)^{m-1}}{m} \sum \frac{[w]}{|w| \prod r_i! s_i!}$ ,  
 424 summed over words  $w = X^{r_1} Y^{s_1} \dots X^{r_m} Y^{s_m}$  with  $r_i + s_i \geq 1$ , where  $[w]$  is right-nested.

- 425 • By submultiplicativity,  $\|[w]\| \leq 2^{|w|-1} r^{|w|}$ .
- 426 • Let  $e_n$  bound  $\|N[w]\|$  for  $|w| = n$ , so  $e_1 = 0$  and  $e_2 \leq \delta r^2$ . Write  $[w] = [W, Z]$  with  $W$  a  
 427 letter.
- 428 •  $[W, P_0 Z]$  has both arguments in  $\mathfrak{a}$ , so its escape is at most  $\delta r \|Z\| \leq \delta 2^{n-2} r^n$ ; and  
 429  $\|[W, N Z]\| \leq 2 r e_{n-1}$ . By induction  $e_n \leq (n-1) 2^{n-2} \delta r^n$ .
- 430 • Each word therefore contributes at most  $\frac{\delta}{4} (2r)^{|w|} / \prod r_i! s_i!$ . The full sum is  $\sum_m \frac{1}{m} (e^{4r} -$   
 431  $1)^m = -\ln(2 - e^{4r})$  for  $r < \ln 2/4$ .
- 432 • The length-1 words lie in  $\mathfrak{a}$  and account for  $4r$ ; subtracting them gives the bound.
- 433 • The length-2 words sum to  $\frac{1}{2} [X, Y]$ , whose escape is at most  $\frac{1}{2} \delta r^2$ ; replacing their crude  
 434 contribution  $4\delta r^2$  gives  $\frac{1}{2} \delta r^2 + O(\delta r^3)$ .
- 435 • The series converges absolutely on this ball, and the principal logarithm is analytic there  
 436 ( $\|e^X e^Y - I\| < 1$ ). Both sides agree near 0, hence on the whole connected ball.

437 (d) With  $d_t = \hat{z}_t - z_t$ :  $d_{t+1} = R_t d_t - e_t$  and  $\|R_t d_t\| = \|d_t\|$ .  $\square$

438 **Proposition 2.** Subalgebras of  $\mathfrak{so}(d)$  carry the ad-invariant negative-definite form  $\text{tr}(XY)$ , hence  
 439 are compact and reductive.

440 (a) A 2-dimensional reductive algebra is abelian. It is maximal abelian because  $\text{rank } \mathfrak{so}(5) = 2$ , and  
 441 maximal abelian subalgebras of a compact Lie algebra are conjugate [Adams, 1969].

442 (b) A 3-dimensional reductive  $\mathfrak{h}$  is its center plus a semisimple part of dimension 0 or 3. It cannot  
 443 be abelian (the rank is 2), so  $\mathfrak{h} \cong \mathfrak{su}(2)$ . Its projections onto the ideals  $\mathfrak{su}(2)_L, \mathfrak{su}(2)_R$  are zero or  
 444 injective, because  $\mathfrak{h}$  is simple.

- 445 • If one projection is zero,  $\mathfrak{h}$  is a factor. The factors are ideals, hence invariant under all of  
 446  $SO(4)$ , and a reflection exchanges them.
- 447 • Otherwise  $\mathfrak{h}$  is the graph of an isomorphism  $\mathfrak{su}(2)_L \rightarrow \mathfrak{su}(2)_R$ . Any two such graphs differ  
 448 by an automorphism of  $\mathfrak{su}(2)$ , which is inner, and  $SU(2) \times SU(2) \rightarrow SO(4)$  is onto, so all  
 449 such  $\mathfrak{h}$  are  $SO(4)$ -conjugate. The stabilizer of  $e_4, \mathfrak{so}(3)$  on  $\mathbb{R}^3 \oplus \mathbb{R}$ , is one of them (it is not  
 450 an ideal).
- 451 •  $SO(4)$  is connected, so it cannot exchange the two ideals; hence three classes under  $SO(4)$   
 452 and, since a reflection fixing  $e_4$  exchanges them and preserves the diagonal, two under  $O(4)$ .

453 The Pfaffian is  $SO(4)$ -invariant and odd under reflections. It vanishes on rank-2 rotations and equals  
 454  $\pm \frac{1}{2}$  on unit elements of  $\mathfrak{su}(2)_{L/R}$ .

(c) A 4-dimensional reductive subalgebra is  $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$  with the  $\mathfrak{u}(1)$  central. The centralizer of the diagonal  $\mathfrak{so}(3)$  is trivial, because  $\mathfrak{so}(4) \cong \Lambda^2 \mathbb{R}^4$  decomposes as  $\mathbf{3} \oplus \mathbf{3}$  under it, with no trivial summand; the same holds for its conjugates. So the  $\mathfrak{su}(2)$  is a factor, which equals the derived algebra  $[\mathfrak{h}, \mathfrak{h}]$ . A diagonal  $\mathfrak{so}(3) \subset \mathfrak{h}$  would satisfy  $\mathfrak{so}(3) = [\mathfrak{so}(3), \mathfrak{so}(3)] \subset [\mathfrak{h}, \mathfrak{h}]$ , a factor, which is impossible.

*Scope.* The classification is for exactly closed spans. Learned spans are only approximately closed; we assign them to classes using continuous invariants (Pfaffian, chirality) and the closure-flow check, without a stability theorem relating approximate closure to distance from a closed class.  $\square$

**Proposition 3.** Here  $g \in SO(3)$  acts on  $z_{1:3}$  and fixes  $z_4$ , so transitions preserve  $\|z\|$ .

(a) Identify  $\mathbb{R}^4$  with the quaternions  $\mathbb{H}$ . The exponential of  $\mathfrak{su}(2)_L$  is left multiplication by unit quaternions, which acts simply transitively on each sphere  $\|z\| = \rho > 0$ . For  $\|z\| = \|z'\|$ ,  $u = z'z^{-1}$  is the unique unit quaternion with  $L_u z = z'$ .

- $q = \log u$  is smooth wherever  $u \neq -1$ , and  $\exp(\log u) = u$  for every branch.
- $u_{02} = z_2 z_0^{-1} = (z_2 z_1^{-1})(z_1 z_0^{-1}) = u_{12} u_{01}$ , so composition is exact.
- Closure is exact, the basis is orthonormal so the Gram penalty is 0, and  $\text{Cov}(z) = I$  in population (states are  $\mathcal{N}(0, I)$  and rotations preserve that law).

The coefficient of a rotation by angle  $\theta = \angle(z, z')$  has entries up to  $\sqrt{2}\theta$ ; only pairs beyond the bound 0.65 break exactness.

(b) Fix the sphere  $S$  and suppose a continuous rule gives  $R(z, z') = \rho(q(z, z')) \in SO(3)$  with  $R(z, z')z = z'$  and  $R(z_0, z_2) = R(z_1, z_2)R(z_0, z_1)$  for all  $z_i \in S$  with pairwise angles  $< \epsilon$ .

- The triple  $(z, z, z)$  gives  $R(z, z) = R(z, z)^2$ , so  $R(z, z) = I$ .
- For a chain  $x_0, \dots, x_n$  with consecutive angles  $< \epsilon/3$ , let  $P = R(x_{n-1}, x_n) \cdots R(x_0, x_1)$ . Inserting or deleting a point inside an  $\epsilon$ -small triangle leaves  $P$  unchanged. Two chains with the same endpoints and pointwise angles  $< \epsilon/3$  are joined by such moves (replace one point at a time; each move is a triangle of diameter  $< \epsilon$ ), so  $P$  is unchanged. Fine subdivisions of a continuous path therefore define one  $P$ , and by uniform continuity it is constant along a homotopy of paths with fixed endpoints.
- $S \cong S^2$  is simply connected, so fixing  $e \in S$ ,  $s(z) = P(e \rightarrow z)$  is well defined, with  $s(z)e = z$ . It is continuous, because  $s(z') = R(z, z')s(z)$  for nearby  $z'$  and  $R$  is continuous with  $R(z, z) = I$ .
- For a unit  $f \perp e$  (in the  $\mathbb{R}^3$  part),  $z \mapsto s(z)f$  is orthogonal to  $s(z)e = z$ : a nowhere-vanishing tangent field on  $S^2$ , contradicting the hairy-ball theorem.

The rule takes values in the true (diagonal) group; the argument uses no property of  $q$  beyond continuity.

(c) For fixed endpoints  $(z, z')$  with  $z_{1:3} \neq 0$ , the set  $\{g : gz = z'\}$  is a coset of the circle stabilizing  $z$ . The states have a density and  $\exp$  pushes the Gaussian coefficients to a law with a density on  $SO(3)$ , so the conditional law of  $g$  on that circle has a density for almost every  $(z, z')$ . Hence  $\mathbb{E}\|\hat{g} - g\|^2 \geq \mathbb{E} \text{tr Cov}(g \mid z, z') > 0$  for every endpoint estimator  $\hat{g}$ .  $\square$

**Remark (what Prop. 3b does not show).**

- *Continuity is essential.*  $S$  minus one point is contractible, so  $SO(3) \rightarrow S^2$  has a continuous section  $s$  there. Ignoring the coefficient bound,  $R(z, z') = s(z')s(z)^{-1}$  is a measurable rule that is exact on every triple avoiding that point, i.e. off a null set.
- *A gap for any fixed Lipschitz class.* Let  $J_\epsilon(q)$  be the expected transport and composition loss on data triples with pairwise angles  $< \epsilon$ , small enough that the chiral rule is within the bound ( $J_\epsilon = 0$  for it). For every  $L$ , the infimum of  $J_\epsilon$  over  $L$ -Lipschitz rules with bounded

coefficients is positive for the true algebra. Otherwise, by Arzelà–Ascoli a subsequence converges locally uniformly to such a rule with  $J_\epsilon = 0$ , by dominated convergence (bounded integrand). The integrand is continuous and the support contains every such triple on every orbit sphere, so the rule is exact there, which contradicts (b).

- *No quantitative statement.* We do not bound this gap, and it may vanish as  $L \rightarrow \infty$ . Nor do we compare the full training objective, which includes large steps and clipping, between the two algebras.

**Corollary 1.** This is Prop. 3(a) applied to both models, together with Prop. 2(b) and the distance  $\frac{1}{2}$  between the diagonal and a chiral span.  $\square$

## B Protocol details

**Architecture and optimization.**  $E, D, q$  are two-hidden-layer SiLU MLPs of width 64. The latent dimension is  $d$ , generators are initialized as  $W \sim \mathcal{N}(0, 1/d)$ , and coefficients are  $0.65 \tanh(\cdot)$ . Training uses AdamW (lr  $2 \times 10^{-3}$ , weight decay  $10^{-5}$ ), batch 160, gradient clipping at 5,  $\epsilon = 10^{-6}$ , Gram weight 1 and anti-collapse weight 16. The anti-collapse weight was chosen in a preliminary round on seeds 5–6 by a method-agnostic criterion. The final iterate is evaluated, with no early stopping. All runs are CPU-only and bit-identical on repeat.

### Order of operations.

1. Selection rules committed.
2. Development grid (280 runs).
3. Rules applied by script.
4. Fresh seeds 100–119 and ablation seeds 200–209, each run once.
5. Review predictions committed.
6. Trace, oracle and rendered runs.

No setting was changed after any fresh-seed result was seen.

**Rendered benchmark.** The image has  $20 \times 20$  pixels on  $[-3.2, 3.2]^2$ . The blob centre is  $(z_1, z_2)$ , its width is  $0.45 e^{0.3z_3}$  and its amplitude is  $1 + 0.5 \tanh z_4$ . Positions are scaled by 0.7 so that blobs stay in view. Centre, width and amplitude determine  $z$ , so the map is injective.

**Oracle intervention.** The transport terms use the true coefficients of each training edge. The composition term is dropped, because  $a_{02}$  is a product rather than a single coefficient.  $q$  is trained only by regression to the true coefficients, for evaluation.

## C Statistical audit

The six primary tests were recomputed from the raw run files by an independent implementation, and all matched. The  $t$ -interval can disagree with the primary sign-flip test when differences are skewed, so we also report the interval obtained by inverting the two-sided exact sign-flip test, which matches the primary procedure, and a percentile bootstrap interval.

**Threshold sensitivity (robustness check, not pre-registered).** SO(3) fresh-seed counts (correct / chiral / other closed / non-closed) under alternative thresholds, as a fraction of the random-span closure level and of the chance distance:

Table 3: Paired differences (BracketNet — +Comp.). Intervals:  $t$ ; sign-flip-inverted; bootstrap.

| Test           | Group | $n$ | median diff. | skew  | $t$ 95%            | inverted sign-flip 95% | bootstrap 95%      |
|----------------|-------|-----|--------------|-------|--------------------|------------------------|--------------------|
| H1 closure     | $T^2$ | 20  | -0.0001      | -1.94 | [-0.5427, +0.0049] | [-0.5331, -0.0067]     | [-0.5410, -0.0380] |
| H2 GT distance | $T^2$ | 20  | -0.0000      | -2.58 | [-0.0674, +0.0240] | [-0.0659, +0.0172]     | [-0.0688, +0.0136] |
| H3 cross-seed  | $T^2$ | 10  | -0.0001      | -0.33 | [-0.1421, +0.1106] | [-0.1410, +0.1091]     | [-0.1217, +0.0862] |
| H1 closure     | SO(3) | 20  | -0.6501      | -0.18 | [-0.8550, -0.3278] | [-0.8566, -0.3269]     | [-0.8295, -0.3572] |
| H2 GT distance | SO(3) | 20  | -0.3321      | 0.21  | [-0.3348, -0.1301] | [-0.3347, -0.1293]     | [-0.3231, -0.1378] |
| H3 cross-seed  | SO(3) | 10  | -0.3466      | 0.30  | [-0.4424, -0.1423] | [-0.4464, -0.1439]     | [-0.4137, -0.1664] |

|  | closed | correct | +Comp.   | BracketNet |
|--|--------|---------|----------|------------|
|  | 0.1%   | 5%      | 3/0/0/17 | 8/5/0/7    |
|  | 0.1%   | 10%     | 3/0/0/17 | 8/5/0/7    |
|  | 0.1%   | 20%     | 3/0/0/17 | 8/5/0/7    |
|  | 1%     | 5%      | 5/0/0/15 | 8/8/0/4    |
|  | 1%     | 10%     | 5/0/0/15 | 8/8/0/4    |
|  | 1%     | 20%     | 5/0/0/15 | 8/8/0/4    |
|  | 5%     | 5%      | 7/0/0/13 | 8/9/1/2    |
|  | 5%     | 10%     | 7/0/0/13 | 9/9/0/2    |
|  | 5%     | 20%     | 7/0/0/13 | 9/9/0/2    |

The qualitative conclusions (no chiral solutions without closure; several with it; more correct-closed runs with closure) hold at every setting.

## D Comparison of settings

| Work                             | Supervision                       | Learned structure              | Closure             | Cross-model structure |
|----------------------------------|-----------------------------------|--------------------------------|---------------------|-----------------------|
| Rao & Ruderman; Dickstein et al. | transformed pairs                 | linear generators              | not enforced        | no                    |
| Cohen & Welling 2014             | pairs, commutative group          | irreps of a torus              | abelian by design   | no                    |
| LieGAN / LaLiGAN                 | distributional invariance         | generators (latent)            | not penalized       | no                    |
| Forestano et al. 2023            | labelled data                     | generators                     | closure loss        | no                    |
| Quessard et al.; Painter et al.  | interaction (Painter: unlabelled) | rotation representations       | SO( $n$ ) by design | no                    |
| Homomorphism AE                  | transitions + action labels       | homomorphism into matrix group | via composition     | no                    |
| CKA / shape metrics              | trained networks                  | —                              | —                   | embeddings only       |
| This work                        | transitions, no labels            | generators + endpoint actions  | closure penalty     | algebras, transfer    |

These summaries describe each setting at the level needed to judge overlap, from abstracts and prior knowledge (the full texts could not be retrieved); see the cited papers for details.

## E Additional results

Table 4: Fresh-seed results for all groups, including SO(2) (closure identically 0 for one generator).

| Group | Method                       | 1-step MSE          | 10-step MSE         | Closure residual                            | GT dist.          | CC    |
|-------|------------------------------|---------------------|---------------------|---------------------------------------------|-------------------|-------|
| SO(2) | Local                        | $0.0026 \pm 0.0010$ | $0.0061 \pm 0.0019$ | $0 \pm 0$                                   | $0.000 \pm 0.000$ | 20/20 |
| SO(2) | +Comp.                       | $0.0026 \pm 0.0010$ | $0.0065 \pm 0.0028$ | $0 \pm 0$                                   | $0.000 \pm 0.000$ | 20/20 |
| SO(2) | BracketNet                   | $0.0026 \pm 0.0010$ | $0.0063 \pm 0.0027$ | $0 \pm 0$                                   | $0.000 \pm 0.000$ | 20/20 |
| SO(2) | BracketNet ( $\lambda=0.1$ ) | $0.0026 \pm 0.0010$ | $0.0063 \pm 0.0027$ | $0 \pm 0$                                   | $0.000 \pm 0.000$ | 20/20 |
| $T^2$ | Local                        | $0.0080 \pm 0.0076$ | $0.0645 \pm 0.0841$ | $3.4 \times 10^{-1} \pm 7.2 \times 10^{-1}$ | $0.161 \pm 0.282$ | 14/20 |
| $T^2$ | +Comp.                       | $0.0081 \pm 0.0076$ | $0.0605 \pm 0.0757$ | $2.9 \times 10^{-1} \pm 6.3 \times 10^{-1}$ | $0.151 \pm 0.274$ | 14/20 |
| $T^2$ | BracketNet                   | $0.0072 \pm 0.0079$ | $0.0539 \pm 0.0792$ | $2.2 \times 10^{-2} \pm 5.6 \times 10^{-2}$ | $0.129 \pm 0.282$ | 16/20 |
| $T^2$ | BracketNet ( $\lambda=0.1$ ) | $0.0082 \pm 0.0078$ | $0.0622 \pm 0.0785$ | $3.1 \times 10^{-1} \pm 6.1 \times 10^{-1}$ | $0.157 \pm 0.277$ | 14/20 |
| SO(3) | Local                        | $0.0048 \pm 0.0027$ | $0.0325 \pm 0.0312$ | $6.3 \times 10^{-1} \pm 5.7 \times 10^{-1}$ | $0.479 \pm 0.435$ | 6/20  |
| SO(3) | +Comp.                       | $0.0048 \pm 0.0031$ | $0.0328 \pm 0.0299$ | $6.2 \times 10^{-1} \pm 5.7 \times 10^{-1}$ | $0.475 \pm 0.433$ | 5/20  |
| SO(3) | BracketNet                   | $0.0049 \pm 0.0034$ | $0.0319 \pm 0.0341$ | $2.5 \times 10^{-2} \pm 7.1 \times 10^{-2}$ | $0.243 \pm 0.254$ | 8/20  |
| SO(3) | BracketNet ( $\lambda=0.1$ ) | $0.0050 \pm 0.0031$ | $0.0484 \pm 0.0484$ | $5.4 \times 10^{-1} \pm 5.4 \times 10^{-1}$ | $0.453 \pm 0.413$ | 5/20  |

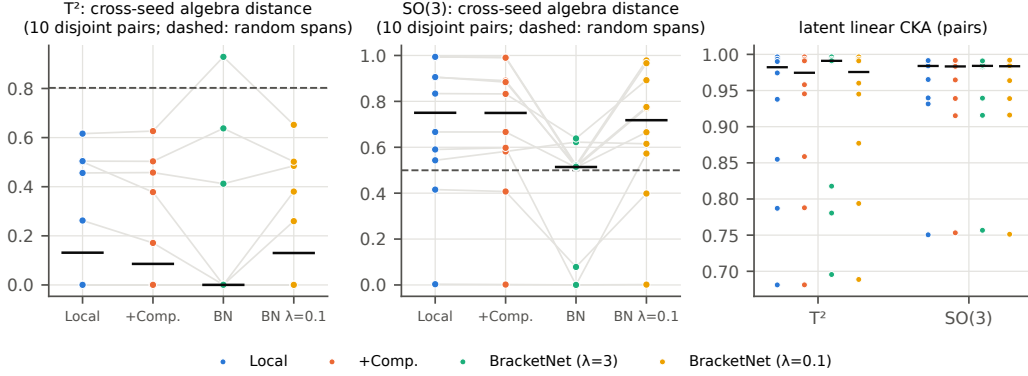


Figure 2: Cross-seed algebra distance on 10 disjoint pairs (grey lines join the same pair across methods; BN = BracketNet) and latent CKA.

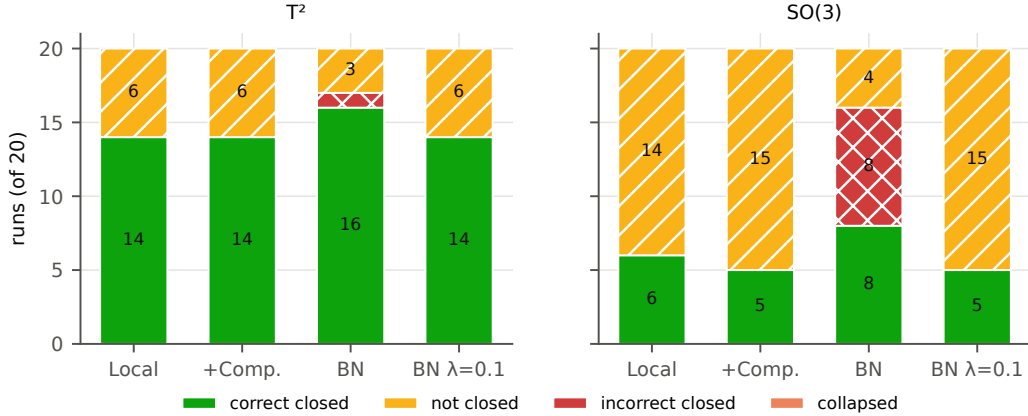


Figure 3: Structural category of each fresh-seed run (BN = BracketNet).

## F Reproducibility

The anonymized supplement contains the data generator, models, training and analysis code, the selection rules and review predictions, per-run result files for every run, the fresh-seed checkpoints,



Table 5: Cross-seed alignment on 10 disjoint pairs (mean $\pm$ s.d.). Algebra distance chance: 0.80 ( $T^2$ ), 0.50 (SO(3)). Transfer gap: 0 = native, 1 = identity. Projected: transferred actions re-expressed in  $b$ 's generators.

| Group | Method                       | CKA               | Algebra dist.     | Action disagr.    | Transfer gap      | Projected gap     |
|-------|------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| $T^2$ | Local                        | 0.920 $\pm$ 0.110 | 0.234 $\pm$ 0.261 | 0.596 $\pm$ 0.662 | 0.876 $\pm$ 1.621 | 0.647 $\pm$ 1.069 |
| $T^2$ | +Comp.                       | 0.920 $\pm$ 0.109 | 0.214 $\pm$ 0.252 | 0.580 $\pm$ 0.659 | 0.852 $\pm$ 1.573 | 0.626 $\pm$ 1.032 |
| $T^2$ | BracketNet                   | 0.925 $\pm$ 0.114 | 0.198 $\pm$ 0.341 | 0.485 $\pm$ 0.785 | 0.769 $\pm$ 1.406 | 0.432 $\pm$ 0.838 |
| $T^2$ | BracketNet ( $\lambda=0.1$ ) | 0.923 $\pm$ 0.106 | 0.228 $\pm$ 0.259 | 0.592 $\pm$ 0.662 | 0.770 $\pm$ 1.398 | 0.611 $\pm$ 0.997 |
| SO(3) | Local                        | 0.951 $\pm$ 0.074 | 0.685 $\pm$ 0.312 | 2.280 $\pm$ 0.995 | 0.745 $\pm$ 1.033 | 0.923 $\pm$ 0.413 |
| SO(3) | +Comp.                       | 0.949 $\pm$ 0.073 | 0.685 $\pm$ 0.308 | 2.222 $\pm$ 0.973 | 0.741 $\pm$ 1.008 | 0.909 $\pm$ 0.393 |
| SO(3) | BracketNet                   | 0.952 $\pm$ 0.073 | 0.392 $\pm$ 0.257 | 2.015 $\pm$ 0.815 | 0.732 $\pm$ 0.919 | 1.257 $\pm$ 0.536 |
| SO(3) | BracketNet ( $\lambda=0.1$ ) | 0.949 $\pm$ 0.074 | 0.664 $\pm$ 0.296 | 2.194 $\pm$ 0.949 | 0.726 $\pm$ 0.978 | 0.924 $\pm$ 0.375 |

Table 6: Structural breakdown of fresh-seed runs (counts out of 20). Incorrect-closed runs are shown with the chiral count in parentheses. “Bad action” counts correct-closed runs whose action disagreement with the truth exceeds 0.5. PR: participation ratio.

| Group | Method                       | Correct | Incorrect (chiral) | Non-closed | Collapsed | Bad action | Action disagr.    | PR              |
|-------|------------------------------|---------|--------------------|------------|-----------|------------|-------------------|-----------------|
| $T^2$ | Local                        | 14      | 0 (0)              | 6          | 0         | 0          | 0.336 $\pm$ 0.438 | 2.98 $\pm$ 0.38 |
| $T^2$ | +Comp.                       | 14      | 0 (0)              | 6          | 0         | 0          | 0.326 $\pm$ 0.429 | 2.93 $\pm$ 0.30 |
| $T^2$ | BracketNet                   | 16      | 1 (0)              | 3          | 0         | 0          | 0.261 $\pm$ 0.405 | 2.86 $\pm$ 0.08 |
| $T^2$ | BracketNet ( $\lambda=0.1$ ) | 14      | 0 (0)              | 6          | 0         | 0          | 0.331 $\pm$ 0.433 | 2.93 $\pm$ 0.26 |
| SO(3) | Local                        | 6       | 0 (0)              | 14         | 0         | 0          | 0.849 $\pm$ 0.355 | 2.26 $\pm$ 0.44 |
| SO(3) | +Comp.                       | 5       | 0 (0)              | 15         | 0         | 0          | 0.838 $\pm$ 0.361 | 2.27 $\pm$ 0.44 |
| SO(3) | BracketNet                   | 8       | 8 (8)              | 4          | 0         | 0          | 0.764 $\pm$ 0.320 | 2.91 $\pm$ 1.00 |
| SO(3) | BracketNet ( $\lambda=0.1$ ) | 5       | 0 (0)              | 15         | 0         | 0          | 0.833 $\pm$ 0.356 | 2.33 $\pm$ 0.46 |

figure scripts and unit tests. Every number in the text is generated as a  $\LaTeX$  macro from the raw files.

## G Use of AI assistance

Code, experiments, analyses and drafts of this manuscript were developed with substantial assistance from an AI coding assistant. The proofs were checked by the same AI system and by automated numerical tests; this self-review is not independent human verification. The authors are responsible for all content.

Table 7: Ablations on seeds 200–209 (mean $\pm$ s.d.). Coverage is the fraction of the true span contained in the learned span. Categories: correct / incorrect / non-closed / collapsed. With misspecified  $K$ , “correct” is impossible by definition.

| Group | Variant                      | 10-step MSE         | Closure                                     | GT dist.          | Coverage          | Categories |
|-------|------------------------------|---------------------|---------------------------------------------|-------------------|-------------------|------------|
| $T^2$ | BracketNet (reference)       | 0.0157 $\pm$ 0.0047 | $3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$ | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 10/0/0/0   |
| $T^2$ | +Comp. (reference)           | 0.0231 $\pm$ 0.0275 | $2.0 \times 10^{-1} \pm 6.2 \times 10^{-1}$ | 0.050 $\pm$ 0.156 | 0.950 $\pm$ 0.156 | 9/0/1/0    |
| $T^2$ | constant penalty, no staging | 0.0370 $\pm$ 0.0475 | $1.2 \times 10^{-4} \pm 2.2 \times 10^{-4}$ | 0.076 $\pm$ 0.170 | 0.924 $\pm$ 0.170 | 8/2/0/0    |
| $T^2$ | staged ramp, no freeze       | 0.0156 $\pm$ 0.0048 | $3.4 \times 10^{-5} \pm 2.1 \times 10^{-5}$ | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 10/0/0/0   |
| $T^2$ | no Gram penalty              | 0.0274 $\pm$ 0.0382 | $1.4 \times 10^{-2} \pm 4.5 \times 10^{-2}$ | 0.027 $\pm$ 0.084 | 0.973 $\pm$ 0.084 | 9/0/1/0    |
| $T^2$ | anti-collapse weight 1       | 0.1997 $\pm$ 0.0253 | $3.1 \times 10^{-3} \pm 4.6 \times 10^{-3}$ | 0.876 $\pm$ 0.130 | 0.124 $\pm$ 0.130 | 0/8/2/0    |
| $T^2$ | span-only (whitened) closure | 0.0157 $\pm$ 0.0047 | $3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$ | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 10/0/0/0   |
| $T^2$ | 0.3 $\lambda$                | 0.0236 $\pm$ 0.0287 | $1.4 \times 10^{-1} \pm 4.5 \times 10^{-1}$ | 0.040 $\pm$ 0.127 | 0.960 $\pm$ 0.127 | 9/0/1/0    |
| $T^2$ | 3 $\lambda$                  | 0.0156 $\pm$ 0.0044 | $1.6 \times 10^{-5} \pm 1.1 \times 10^{-5}$ | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 10/0/0/0   |
| $T^2$ | $K+1$ generators             | 0.0248 $\pm$ 0.0194 | $3.2 \times 10^{-1} \pm 1.1 \times 10^{-1}$ | 0.375 $\pm$ 0.103 | 0.937 $\pm$ 0.154 | 0/1/9/0    |
| $T^2$ | +Comp., $K+1$                | 0.0222 $\pm$ 0.0245 | $8.2 \times 10^{-1} \pm 1.9 \times 10^{-1}$ | 0.367 $\pm$ 0.108 | 0.949 $\pm$ 0.161 | 0/0/10/0   |
| $T^2$ | $K-1$ generators             | 0.1917 $\pm$ 0.0300 | 0 $\pm$ 0                                   | 0.501 $\pm$ 0.001 | 0.499 $\pm$ 0.001 | 0/10/0/0   |
| SO(3) | BracketNet (reference)       | 0.0218 $\pm$ 0.0160 | $4.2 \times 10^{-2} \pm 8.0 \times 10^{-2}$ | 0.244 $\pm$ 0.298 | 0.756 $\pm$ 0.298 | 5/2/3/0    |
| SO(3) | +Comp. (reference)           | 0.0393 $\pm$ 0.0309 | $7.0 \times 10^{-1} \pm 5.2 \times 10^{-1}$ | 0.462 $\pm$ 0.412 | 0.538 $\pm$ 0.412 | 3/0/7/0    |
| SO(3) | constant penalty, no staging | 0.0298 $\pm$ 0.0394 | $1.4 \times 10^{-2} \pm 2.6 \times 10^{-2}$ | 0.185 $\pm$ 0.298 | 0.815 $\pm$ 0.298 | 6/1/3/0    |
| SO(3) | staged ramp, no freeze       | 0.0299 $\pm$ 0.0191 | $2.6 \times 10^{-2} \pm 4.6 \times 10^{-2}$ | 0.239 $\pm$ 0.301 | 0.761 $\pm$ 0.301 | 5/2/3/0    |
| SO(3) | no Gram penalty              | 0.0181 $\pm$ 0.0136 | $1.5 \times 10^{-2} \pm 4.3 \times 10^{-2}$ | 0.177 $\pm$ 0.260 | 0.823 $\pm$ 0.260 | 6/2/2/0    |
| SO(3) | anti-collapse weight 1       | 0.0302 $\pm$ 0.0140 | $5.6 \times 10^{-2} \pm 2.2 \times 10^{-2}$ | 0.619 $\pm$ 0.063 | 0.381 $\pm$ 0.063 | 0/1/9/0    |
| SO(3) | span-only (whitened) closure | 0.0182 $\pm$ 0.0138 | $4.0 \times 10^{-2} \pm 7.7 \times 10^{-2}$ | 0.243 $\pm$ 0.298 | 0.757 $\pm$ 0.298 | 5/2/3/0    |
| SO(3) | 0.3 $\lambda$                | 0.0314 $\pm$ 0.0284 | $2.2 \times 10^{-1} \pm 3.4 \times 10^{-1}$ | 0.299 $\pm$ 0.297 | 0.701 $\pm$ 0.297 | 4/1/5/0    |
| SO(3) | 3 $\lambda$                  | 0.0274 $\pm$ 0.0244 | $1.7 \times 10^{-2} \pm 3.0 \times 10^{-2}$ | 0.237 $\pm$ 0.302 | 0.763 $\pm$ 0.302 | 5/2/3/0    |
| SO(3) | $K+1$ generators             | 0.0225 $\pm$ 0.0108 | $1.2 \times 10^{-1} \pm 2.4 \times 10^{-1}$ | 0.457 $\pm$ 0.076 | 0.723 $\pm$ 0.102 | 0/7/3/0    |
| SO(3) | +Comp., $K+1$                | 0.0156 $\pm$ 0.0087 | $5.5 \times 10^{-1} \pm 2.6 \times 10^{-1}$ | 0.511 $\pm$ 0.220 | 0.652 $\pm$ 0.293 | 0/0/10/0   |
| SO(3) | $K-1$ generators             | 0.1253 $\pm$ 0.0535 | $6.0 \times 10^{-1} \pm 5.1 \times 10^{-1}$ | 0.464 $\pm$ 0.166 | 0.536 $\pm$ 0.166 | 0/4/6/0    |

Table 8: Development grid (seeds 5–9; selection only; means over 5 seeds). For each method: 10-step MSE, closure, correct-closed count.

| Group | Updates | MSE    | Local closure | CC  | MSE    | +Comp. closure | CC  | MSE    | BracketNet closure | CC  |
|-------|---------|--------|---------------|-----|--------|----------------|-----|--------|--------------------|-----|
| $T^2$ | 420     | 0.1635 | 0.1374        | 1/5 | 0.1604 | 0.1104         | 1/5 | 0.1647 | 0.0600             | 1/5 |
| $T^2$ | 1000    | 0.0704 | 0.0223        | 4/5 | 0.0678 | 0.0188         | 4/5 | 0.0676 | 0.0011             | 4/5 |
| $T^2$ | 2000    | 0.0376 | 0.0016        | 4/5 | 0.0391 | 0.0022         | 4/5 | 0.0390 | 0.0005             | 4/5 |
| $T^2$ | 4000    | 0.0308 | 0.0004        | 4/5 | 0.0312 | 0.0003         | 4/5 | 0.0317 | 0.0001             | 4/5 |
| SO(3) | 420     | 0.1391 | 0.8088        | 0/5 | 0.1532 | 0.7579         | 0/5 | 0.1479 | 0.3826             | 0/5 |
| SO(3) | 1000    | 0.0713 | 0.8762        | 1/5 | 0.0843 | 0.7771         | 1/5 | 0.0668 | 0.2032             | 1/5 |
| SO(3) | 2000    | 0.0396 | 0.9245        | 1/5 | 0.0308 | 0.6989         | 1/5 | 0.0573 | 0.1129             | 2/5 |
| SO(3) | 4000    | 0.0295 | 0.9574        | 1/5 | 0.0276 | 0.7086         | 1/5 | 0.0340 | 0.0568             | 2/5 |

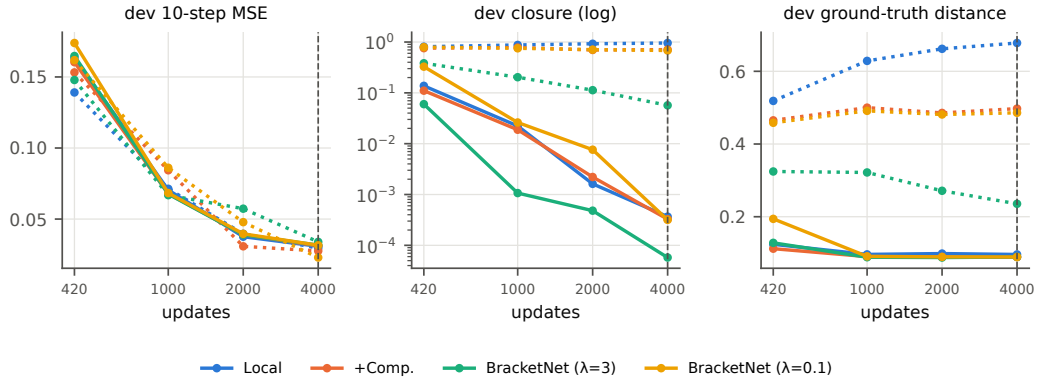


Figure 4: Development-seed budget sweep (solid:  $T^2$ ; dotted: SO(3)); the vertical line marks the selected budget.

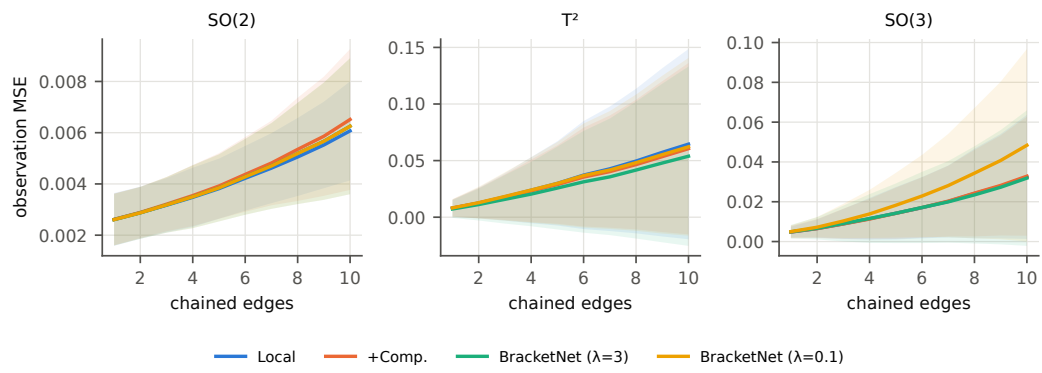


Figure 5: Chained transport error (mean  $\pm$  s.d., 20 fresh seeds).